

1 EX-001 — extruder assembly

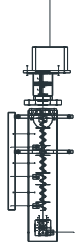
물리 검증·안전 인증·통전 승인이 아니다. E-stop, lid/service interlock, branch fuse, 독립 thermal fuse를 정상 firmware와 독립 구현하고 exact received component 정격·배선·보호소자를 실측 확인하기 전 통전하지 않는다.

Revision: final-design-fabrication-closure-v0.8 · 상태: DIGITAL_DOCUMENT / PHYSICAL_NOT_RUN / USER_APPROVAL_REQUIRED

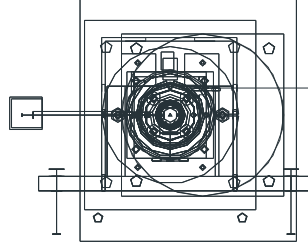
EX-001 extruder assembly

v0.8 · mm · 제3각법 · source: GGM integrated FreeCAD assembly · bodies 53

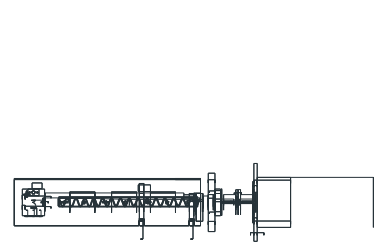
TOP-X-Y



FRONT X-Z



SIDE Y-Z



Overall X 168.40 · Y 646.00 · Z 140.00 mm · 치수/공차는 개별 제작도면과 interface catalog 우선
Revision final-design-fabrication-closure-v0.8 · physical validation NOT_RUN

1.1 제작·검사 기준

- 단위: mm · 제3각법 · 축척: NTS(기입 치수 우선)
- 재료/구성: SCM440 screw/barrel / steel supports
- 핵심 치수/공차: integral barrel shoulder + rear retainer; $2 \times M4 \times 25$ class8.8 at 2.9 N·m; cold endplay 0.12–0.28 mm; front guide axial travel ≥ 1.50 mm; mount $M5 \times 4 = 2.5$ N·m
- 일반공차: ISO 2768-m. 개별 부품은 exports/final/manufacturing 도면과 interface_catalog.csv를 함께 검사한다.
- Source: c5bc6892f38af0b79a2e8f7c0b5c01323b536b37 · 물리 검증: NOT_RUN